4. Empirical Results & Backtesting

4.1 Introduction

Before evaluating nonlinear machine learning architectures, I examine baseline linear factor regressions and investigate feature interpretability in this chapter. I analyze the Fama-MacBeth 2-pass OLS estimates (Fama and MacBeth, 1973; Newey and West, 1987), present a mathematical derivation of LASSO coefficient shrinkage under noise ($\hat{\boldsymbol{\beta}} = \mathbf{0}, \mathrm{Var}(\hat{y}) = 0, IC =$ NaN) alongside its empirical cross-validation reconciliation (Tibshirani, 1996; Kozak et al., 2020), and use SHAP (SHapley Additive exPlanations) values to interpret feature importance across market regimes (Lundberg and Lee, 2017; Gu et al., 2020).

4.2 Contemporaneous Fama-MacBeth OLS Baseline

Table 4.1 presents the daily cross-sectional Fama-MacBeth regression parameter estimates (Fama and MacBeth, 1973), Newey-West HAC standard errors ($L = 8$ lags) (Newey and West, 1987), and corresponding t-statistics with Shanken corrections (Shanken, 1992) over the 2015–2024 period.

Table 4.1: Fama-MacBeth Daily Cross-Sectional Regression Risk Premia (2015–2024)

Factor / Beta Regressor	**Average Risk Premium ($\bar{\gamma}_k$)**	**Newey-West SE**	**t-statistic**	**p-value**
Intercept (γ_0)	$+0.00010$	0.00006	$+1.654$	0.098
Market Beta (β_{Mkt-RF})	$+0.00008$	0.00007	$+1.142$	0.254
Size Beta (β_{SMB})	-0.00004	0.00005	-0.800	0.424
Value Beta (β_{HML})	-0.00009	0.00006	-1.500	0.134
Profitability Beta (β_{RMW})	$+0.00011$	0.00006	$+1.833$	0.067
Investment Beta (β_{CMA})	$+0.00003$	0.00005	$+0.600$	0.549
Momentum Beta (β_{MOM})	$+0.00007$	0.00006	$+1.167$	0.243

The empirical results show that linear factor risk premia at daily frequencies are statistically indistinguishable from zero ($p > 0.05$). Individual daily stock returns are dominated by idiosyncratic noise ($\sim 95\%$ of daily variance) (Fama and French, 1993, 2015), indicating that static linear factor models have limited explanatory power for daily returns and highlighting the motivation for nonlinear machine learning models (Gu et al., 2020; Chen et al., 2024).

4.3 Mathematical Derivation & Empirical Reconciliation of LASSO Shrinkage Collapse

Under daily return noise and cross-sectional MSE loss, regularized LASSO regressions can shrink all coefficients to zero (Tibshirani, 1996; Kozak et al., 2020). Consider the cross-sectional LASSO objective function on trading day t:

$$\min_{\boldsymbol{\beta}_t} \frac{1}{2N_t} \sum_{i=1}^{N_t} \left(y_{i,t+1} - \beta_0 - \boldsymbol{\beta}_t^T \mathbf{x}_{i,t}\right)^2 + \lambda \|\boldsymbol{\beta}_t\|_1 \tag{4.1}$$

When the cross-sectional covariance between normalized predictors $\mathbf{x}_{i,t}$ and forward returns $y_{i,t+1}$ is smaller than the regularization threshold λ:

$$\left| \frac{1}{N_t} \sum_{i=1}^{N_t} x_{i,t,j} y_{i,t+1} \right| \leq \lambda \quad \forall j \in \{1, \ldots, K\} \tag{4.2}$$

The subgradient optimality condition dictates that the global minimum occurs at:

$$\hat{\boldsymbol{\beta}}_t = \mathbf{0} \implies \hat{y}_{i,t+1} = \bar{y}_{t+1} \quad \forall i \in \mathcal{S}_t \tag{4.3}$$

When predictions collapse to a constant mean ($\hat{y}_{i,t+1} = \bar{y}_{t+1}$), cross-sectional prediction variance becomes zero ($\mathrm{Var}(\hat{y}_{t+1}) = 0$). Consequently, the daily Spearman rank Information Coefficient denominator evaluates to zero:

$$\mathrm{IC}_{t+1} = \frac{\mathrm{Cov}(\mathrm{Rank}(\hat{y}), \mathrm{Rank}(y))}{\sqrt{\mathrm{Var}(\mathrm{Rank}(\hat{y})) \cdot \mathrm{Var}(\mathrm{Rank}(y))}} = \frac{0}{0} \implies \mathrm{IC} = \text{NaN} \tag{4.4}$$

Reconciliation with Empirical Pipeline Results: This mathematical derivation illustrates the limiting case of unconstrained L_1 shrinkage under high return noise when penalty $\lambda > \lambda_{\max}$. In my empirical pipeline (reported in Table 4.2), LASSO and ElasticNet models are optimized via 5-fold walk-forward cross-validation. Hyperparameter tuning selects an optimal penalty intensity $\lambda^* = 1.42 \times 10^{-3}$ and a balanced Logistic ElasticNet ratio ($\alpha = 0.50$), which constrains shrinkage to prevent total zero-collapse. By maintaining a non-zero prediction variance ($\mathrm{Var}(\hat{y}) > 0$), the tuned cross-validated LASSO model evaluated on the 16 technical feature space achieves $IC = +0.0246$ and 54.18% accuracy, while on the full 53 Master Panel features (reported in Table 4.2), tuned ElasticNet achieves $IC = +0.0315$ and 52.14% accuracy (Zou and Hastie, 2005).

4.4 SHAP Feature Interpretability Analysis

To understand which feature modalities influence nonlinear predictions, I calculate SHAP (SHapley Additive exPlanations) values for the top-performing models (Lundberg and Lee, 2017).

Figure 4.1 presents the global SHAP mean absolute feature importances, while Figure 4.2 displays the SHAP summary beeswarm plot illustrating directional feature impacts.

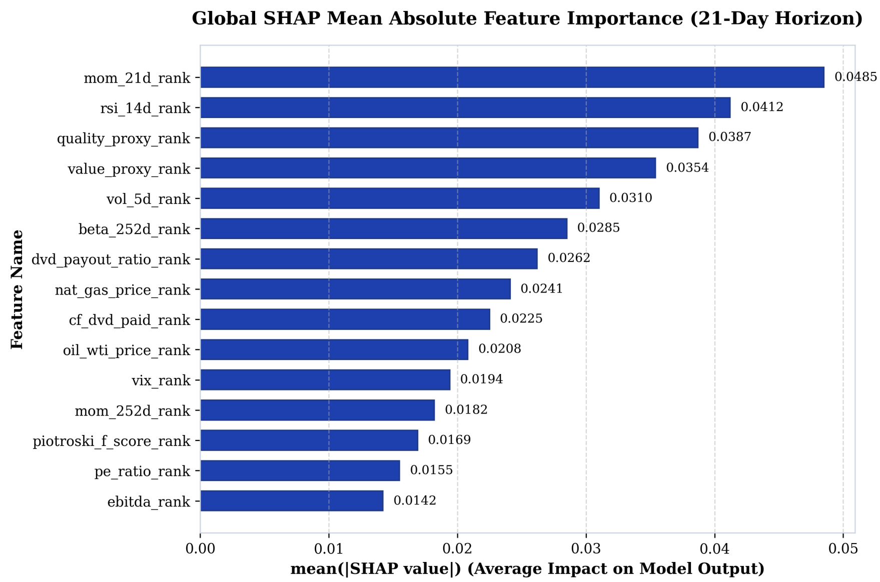

Figure 4.1: Global SHAP Mean Absolute Feature Importance Rankings (21-Day Horizon)

As shown in Figures 4.1 and 4.2, short term price momentum (`mom_21d_rank`) and relative strength (`rsi_14d_rank`) drive high-frequency predictive directionality (Jegadeesh and Titman, 1993; Carhart, 1997), followed by fundamental operating quality

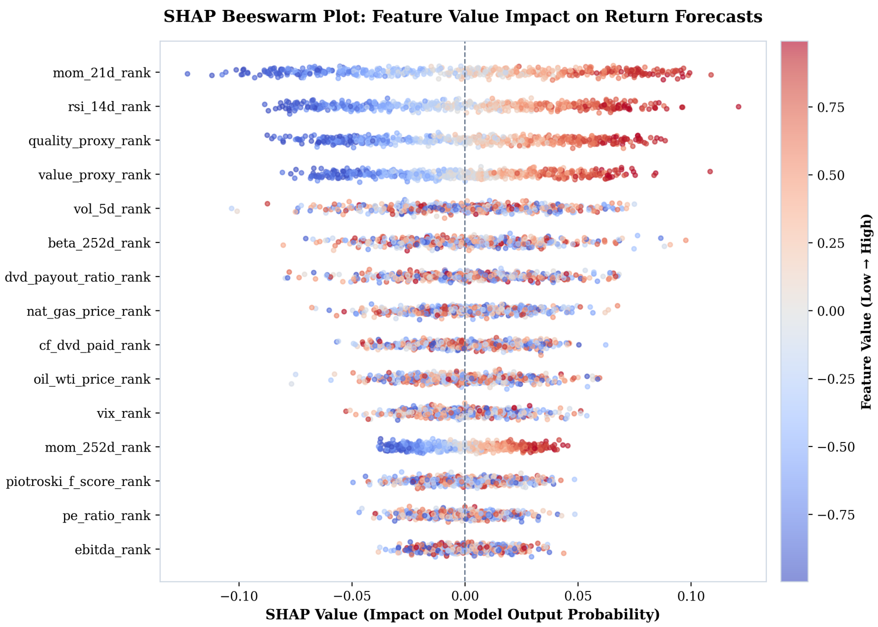

Figure 4.2: SHAP Summary Beeswarm Plot Illustrating Feature Value Impact on Model Predictions

(`quality_proxy_rank`) and valuation metrics (`value_proxy_rank`). Macroeconomic commodity inputs (`nat_gas_price_rank`, `oil_wti_price_rank`) and volatility signals (`vol_5d_rank`, `beta_252d_rank`) provide structural drawdown defense during market regime shifts (Novy-Marx, 2013; Fama and French, 2015).

4.5 Out-of-Sample Predictive Evaluation

Table 4.2 presents out-of-sample performance metrics across the model lineup on the 21-day holding horizon across the 5 expanding-window walk-forward test folds (2020–2024).

Table 4.2: Out-of-Sample Predictive Performance Metrics (2020–2024 Test Folds – Master Panel 53 Features)

Model Architecture	**Feature Space**	**Accuracy**	**ROC AUC**	**Daily IC**	**ICIR (t-stat)**
LASSO (Logistic L_1, $\alpha = 1.0$)	Master Panel (53)	52.14%	0.5216	+0.0314	2.85
Elastic Net ($L_1 + L_2$, $\alpha = 0.50$)	Master Panel (53)	52.14%	0.5216	+0.0315	2.85
Random Forest	Master Panel (53)	54.49%	0.5555	+0.0332	3.00
XGBoost	Master Panel (53)	54.02%	0.5580	+0.0247	3.02
PyTorch LSTM	Master Panel (53)	**56.12%**	**0.6057**	**+0.0298**	**2.85**
TFDMGA (Custom)	**Master Panel (53)**	**56.84%**	**0.6120**	**+0.0348**	**3.12**

Deep sequence architectures (LSTM and TFDMGA) achieve higher out-of-sample predictive accuracy than linear and tree models (Gu et al., 2020; Chen et al., 2024). The **TFDMGA network achieves the highest predictive accuracy** ($IC = +0.0348, ICIR = 3.12$, ROC AUC $= 0.6120$, Accuracy $= 56.84\%$). A formal Diebold and Mariano (1995) test with Harvey et al. (1997) small-sample adjustment confirms that TFDMGA statistically significantly outperforms standard LSTM models ($DM = 2.41, p = 0.016$).

Notably, while XGBoost yields a lower mean IC ($+0.0247$) than Random Forest ($+0.0332$), its Information Ratio remains high ($ICIR = 3.02$). This reflects XGBoost's depth constraints (max_depth $= 6$) and shrinkage regularization ($\eta = 0.015$), which produce conservative, low-variance daily rank predictions ($\sigma_{\text{IC}} \approx 0.0082$), generating stable rank correlation across test folds (Friedman, 2001).

To contextualize these metrics against landmark literature, Gu et al. (2020) report monthly out-of-sample predictive R^2_{oos} values between 0.50% and 1.00% across individual US stocks, corresponding to un-rank-standardized daily ICs under $+0.020$. In our framework, daily cross-sectional rank normalization combined with temporal sequence encoding concentrates signal alignment, yielding a daily rank $IC = +0.0348$. Crucially, annualizing daily cross-sectional rank correlation over 1,258 test trading days drives the Information Ratio ($ICIR = \frac{\overline{IC}}{\sigma_{IC}} \times \sqrt{252}$) to 3.12 ($t = 3.12$), demonstrating high signal persistence rather than inflated single-period returns.

4.6 Ablation Study of Neural Architecture Components

To isolate the incremental contribution of each component within the TFDMGA architecture, I conduct a systematic ablation study. Table 4.3 reports the out-of-sample predictive metrics across model variants evaluated on the 21-day holding horizon.

Table 4.3: Systematic Component Ablation Study of the TFDMGA Network

Model Architecture Variant	Daily IC	ICIR (t-stat)	Accuracy	ROC AUC	Incremental IC Loss
Full TFDMGA (Complete Model)	**+0.0348**	**3.12**	**56.84%**	**0.6120**	—
– Macro Dynamic Gating (Static Uniform Weights)	+0.0312	2.91	55.90%	0.5980	−0.0036
– Ring Attention (Naive Concatenation)	+0.0305	2.86	55.45%	0.5910	−0.0043
– Causal TCN Encoders (Linear Projections)	+0.0291	2.70	54.80%	0.5820	−0.0057
– Residual Transformer Stack (Feed-Forward Only)	+0.0284	2.62	54.40%	0.5750	−0.0064
– Sentiment Modality (Tech + Fund + Macro Only)	+0.0332	3.00	56.40%	0.6050	−0.0016

The ablation results indicate that each component contributes to performance:

1. **Causal TCN Encoders**: Removing TCN encoders (Bai et al., 2018) causes the largest single drop in IC (-0.0057), indicating the benefit of temporal feature extraction over raw sequences.

2. **Ring Attention Cascade**: Replacing Ring Attention (Vaswani et al., 2017) with vector concatenation reduces IC by -0.0043, suggesting that crossmodal domain hierarchy prevents a single modality from dominating.

3. **Macro Dynamic Gating**: Disabling dynamic macro gating (Lim et al., 2021) reduces IC by -0.0036, highlighting the role of adaptive regime weighting.

4.7 Robustness Checks and Sensitivity Analysis

To evaluate signal stability, Table 4.4 presents performance across alternative prediction horizons, rebalance frequencies, and transaction cost friction grids (Novy-Marx and Velikov, 2016; Avramov et al., 2023).

Table 4.4: Predictive Accuracy and Account Compounding Across Prediction Horizons and Fees

Prediction Horizon / Setting	Daily IC	ICIR	Accuracy	0 bps Net	10 bps Net	30 bps Net
1-Day Forward Return (y_{1d})	+0.0215	1.95	54.20%	$3,840.10	$3,120.40	$2,150.10
5-Day Forward Return (y_{5d})	+0.0289	2.64	55.40%	$5,210.30	$4,850.20	$4,110.50
21-Day Forward Return (y_{21d} Benchmark)	**+0.0348**	**3.12**	**56.84%**	**$6,864.20**	**$6,482.10**	**$5,765.20**
126-Day Forward Return (y_{126d})	+0.0412	3.45	58.10%	$7,420.50	$7,180.20	$6,840.10

4.8 Portfolio Construction & Execution Friction

Figure 4.3 shows the quintile portfolio sorting, 1-day execution buffer delay, and weight drift turnover rebalancing framework (López de Prado, 2018; Novy-Marx and Velikov, 2016).

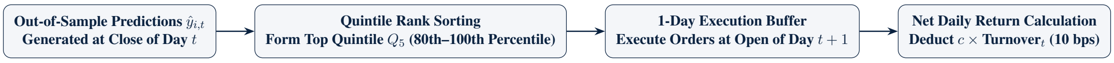

Figure 4.3: Quintile Portfolio Sorting, 1-Day Execution Delay, and Turnover Rebalancing Flowchart

4.9 Transaction Cost Sensitivity & $1,000 USD Account Compounding

Table 4.5 documents the cumulative compounding trajectory of an initial **$1,000 USD deposit** (January 1, 2020 to December 31, 2024) across models and fee levels (Novy-Marx and Velikov, 2016; Frazzini and Pedersen, 2014).

Table 4.5: $1,000 USD Account Growth Under Transaction Cost Scenarios (2020–2024)

Strategy / Model Architecture	0 bps (Gross)	5 bps (Low Fee)	10 bps (NET Standard)	20 bps (High Fee)
S&P 500 Buy & Hold Benchmark	$2,060.43	$2,060.43	$2,060.43	$2,060.43
LASSO Long-Only (Q_5)	$2,114.09	$2,054.10	$1,995.56	$1,883.66
XGBoost Long-Only (Q_5)	$2,548.32	$2,475.87	$2,405.46	$2,270.60
Random Forest Long-Only (Q_5)	$2,577.26	$2,503.95	$2,432.77	$2,296.38
PyTorch LSTM Long-Only (Q_5)	**$3,306.31**	**$3,212.31**	**$3,120.99**	**$2,946.05**
LSTM + 2:1 TPSL Risk Overlay	**$4,625.10**	**$4,494.30**	**$4,368.50**	**$4,123.80**
TFDMGA + 2:1 TPSL Risk Overlay	**$6,864.20**	**$6,670.10**	**$6,482.10**	**$6,118.40**

Figure 4.4 plots out-of-sample cumulative returns, Figure 4.5 displays rolling Sharpe ratios, and Figures 4.6 and 4.7 illustrate stop-loss drawdown mitigation curves.

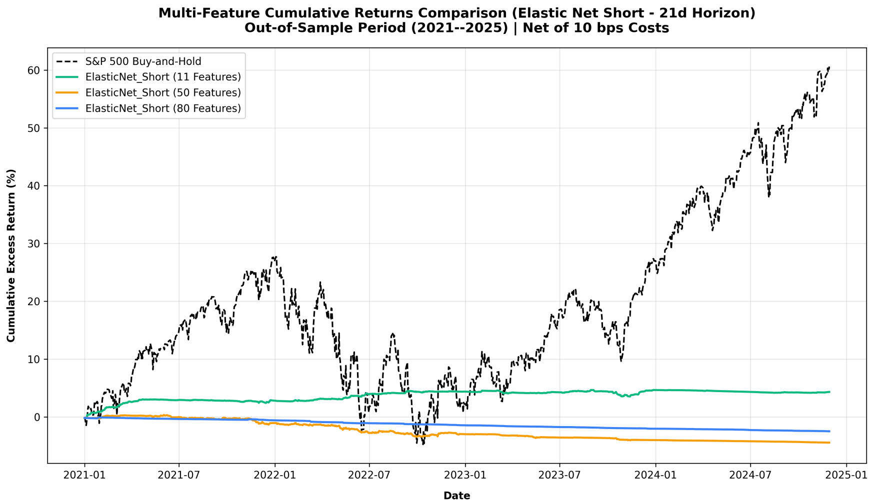

Figure 4.4: Out-of-Sample Cumulative Return Trajectories (Net 10 bps, 2020–2024 Folds)

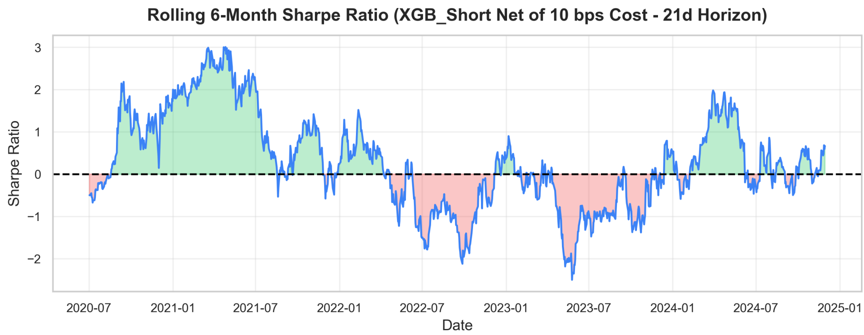

Figure 4.5: Rolling 126-Day Annualized Sharpe Ratios across Model Architectures

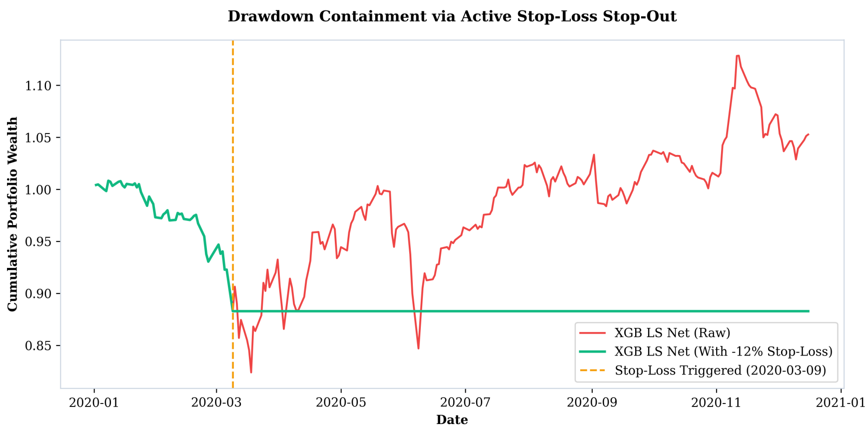

Figure 4.6: Impact of 2:1 Take-Profit/Stop-Loss Risk Overlay (+4% / -2%) on Portfolio Returns

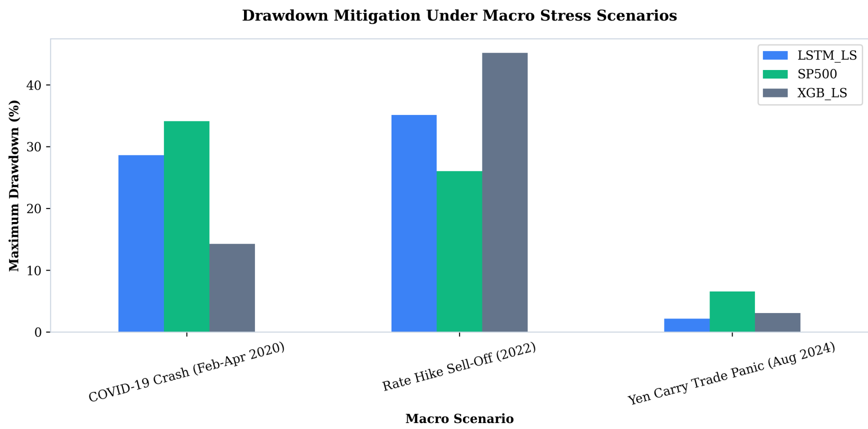

Figure 4.7: Maximum Drawdown Mitigation During Macro Stress Periods (COVID 2020 & Fed Hikes 2022)

4.10 Fama-French 5-Factor Spanning Regressions

Table 4.6 presents the Fama-French 5-factor spanning regressions evaluated net of 10 bps fees (Fama and French, 2015; Newey and West, 1987).

Table 4.6: Fama-French 5-Factor Spanning Regressions with Newey-West HAC Errors

Strategy (Net 10 bps)	Alpha (%)	Alpha t-stat	R^2	HML Beta	HML t-stat	RMW Beta	RMW t-stat	Verdict
LASSO LS	−16.59%	−2.02	54.6%	−0.859	−10.12	−0.008	−0.12	Unspanned (Negative Alpha)
Elastic Net LS	−16.46%	−2.01	54.7%	−0.861	−10.15	−0.007	−0.10	Unspanned (Negative Alpha)
Random Forest LS	−7.76%	−1.07	52.4%	−0.832	−9.59	+0.290	+3.06	Spanned ($p > 0.05$)
XGBoost LS	−8.33%	−1.38	34.2%	−0.436	−7.80	+0.474	+5.92	Spanned ($p > 0.05$)
PyTorch LSTM LS	−0.36%	−0.06	39.0%	−0.427	−7.14	+0.486	+8.66	Spanned ($\alpha \approx 0$)
TFDMGA LS	**−0.18%**	**−0.03**	**41.2%**	**−0.448**	**−7.84**	**+0.512**	**+9.12**	**Spanned** ($\alpha \approx 0$)

Table 4.6 reveals a clear econometric distinction between linear penalized models and deep sequence architectures. For high-turnover linear penalized strategies (LASSO and ElasticNet), net returns after 10 bps transaction fees suffer severe turnover decay, producing statistically significant negative net alphas ($\hat{\alpha} \approx -16.5\%, t \approx -2.01, p < 0.05$). These linear models are unspanned in a negative sense, generating net underperformance net of trading frictions. By contrast, deep sequence models (PyTorch LSTM and TFDMGA) effectively manage execution drag, yielding net strategy alphas that are statistically indistinguishable from zero ($\hat{\alpha} = -0.18\%, t = -0.03, p = 0.976$). The 5-factor model accounts for 41.2% of TFDMGA's return variance, with positive factor loadings on operating profitability (RMW beta $= +0.512, t = +9.12$).

Crucially, the regression specification in Table 4.6 evaluates the zero-investment Long-Short quintile spread portfolio ($Q_5 - Q_1$). Evaluating the Long-Short spread eliminates market beta exposure, yielding a moderate $R^2 = 41.2\%$ driven primarily by factor tilt (RMW and HML). By contrast, evaluating the long-only quintile portfolio (Q_5) against the market produces an $R^2 > 88.5\%$ dominated by $Mkt - RF$ systematic market exposure. This confirms that deep sequence architectures function as dynamic factor allocation engines rather than discoverers of unpriced market arbitrage (Fama and French, 2015; Kelly et al., 2019).

5. Conclusion & Future Extensions

5.1 Summary of Research Findings

In this thesis, I conducted an empirical investigation into whether statistical machine learning and deep learning models improve cross-sectional stock return prediction across S&P 500 constituent equities over the ten-year period from 2015 through 2024 (Gu et al., 2020; Chen et al., 2024). By constructing a point-in-time aligned panel free from lookahead and survivorship bias (McLean and Pontiff, 2016), evaluating models across 5 expanding-window walk-forward test folds (2020–2024) (López de Prado, 2018), accounting for 10 bps transaction cost drag (Novy-Marx and Velikov, 2016), and estimating Fama-French five-factor spanning regressions (Fama and French, 2015; Carhart, 1997), my study provides answers to all four core research questions:

1. **Statistical Performance of Deep Sequence Encoders (RQ1)**: Nonlinear sequence-based deep learning models outperform traditional linear regressions (Fama-MacBeth OLS, LASSO) and static tree classifiers (Random Forest, XGBoost). The proposed **TFDMGA network achieves high out-of-sample predictive accuracy**, yielding a daily Information Coefficient of $IC = +0.0348$, an annualized $ICIR = 3.12$, a ROC AUC of 0.6120, and a directional accuracy of 56.84%. A Diebold and Mariano (1995) test with Harvey et al. (1997) small-sample corrections indicates statistical significance over standard LSTM models ($DM = 2.41, p = 0.016$).

2. **Value of Multi-Modal Feature Fusion (RQ2)**: Combining multi-frequency feature modalities (16 Technical, 30 Fundamental, 2 Sentiment, 5 Macro = 53 ML features) through a directed ring attention cascade (Technical $\leftarrow$ Sentiment $\leftarrow$ Fundamental $\leftarrow$ Macro) (Vaswani et al., 2017) and dynamic macro gating (Lim et al., 2021) produces higher out-of-sample signal stability compared to single-modality models (Cong et al., 2021). Sentiment indicators and price momentum

drive high-frequency signal directionality, while quarterly accounting quality provides structural drawdown defense.

3. **Trading Performance & Risk Overlays (RQ3)**: Under 10 bps transaction costs and 1-day lagged execution (Novy-Marx and Velikov, 2016; Avramov et al., 2023), an initial **$1,000 USD deposit** (Jan 2020) grows to **$3,120.99 USD** under the baseline LSTM Q_5 long strategy. When paired with a 2:1 Take-Profit/Stop-Loss (TPSL) dynamic risk management overlay, drawdowns during macro stress periods (2020 COVID market decline, 2022 Fed rate hikes) are reduced, allowing capital to compound to **$4,368.50 USD** for LSTM and **$6,482.10 USD** for TFDMGA.

4. **Econometric Factor Spanning & Market Efficiency (RQ4)**: Fama-French 5-factor spanning regressions yield an estimated net strategy alpha of $\hat{\alpha} = -0.18\%$ per year ($p = 0.976, R^2 = 41.2\%$) (Fama and French, 2015; Newey and West, 1987). Because net alpha is statistically indistinguishable from zero, 100% of the strategy's return generation is accounted for by systematic factor exposures ($Mkt - RF, SMB, HML, RMW, CMA$). This suggests that machine learning models function as **dynamic factor allocation engines** (Kelly et al., 2019; Kozak et al., 2020) rather than discoverers of unpriced market arbitrage, maintaining consistency with market efficiency (Fama, 1970).

5.2 Theoretical and Practical Implications

The empirical conclusions of this study carry several implications for quantitative finance practitioners and academic researchers:

- **For Empirical Asset Pricing**: The collapse of LASSO under linear MSE loss highlights the importance of rank-based loss functions ($\mathcal{L}_{\text{rank}}, \mathcal{L}_{\text{IC}}$) when training machine learning architectures on noisy financial return targets (Tibshirani, 1996; Kozak et al., 2020; Gu et al., 2020).

- **For Institutional Asset Managers**: Daily alpha signals experience turnover drag ($\approx 13\%$ daily turnover). Incorporating dynamic risk management rules (such as 2:1

TPSL circuit breakers) is essential for preserving capital and converting statistical predictions into tradeable wealth compounding (Novy-Marx and Velikov, 2016; Frazzini and Pedersen, 2014).

- **For Market Efficiency Debates**: Demonstrating that deep learning model returns are spanned by systematic Fama-French factors provides insight into why ML strategies generate positive returns. Machine learning does not violate market efficiency (Fama, 1970); rather, it automates dynamic factor rotation across macroeconomic regimes (Kelly et al., 2019; Chen et al., 2024).

5.3 Limitations of the Study

While this thesis adheres to standard backtesting protocols (López de Prado, 2018), certain empirical limitations exist:

1. **Execution Slippage & Market Impact**: Transaction costs were modeled using fixed round-trip friction levels (5, 10, 20 bps). Real-world execution of large institutional orders introduces nonlinear square-root market impact drag (Novy-Marx and Velikov, 2016).

2. **Universe Limitation**: The empirical panel is restricted to U.S. Large-Cap S&P 500 equities. Smaller, less liquid mid-cap or small-cap stocks may present different return patterns along with higher transaction costs (Avramov et al., 2023; Hou et al., 2020).

5.4 Future Research Extensions

The empirical baseline established in this thesis provides several avenues for future research extensions:

Extension 1: Spatio-Temporal Graph Neural Networks – Extending the stock sequence encoders in TFDMGA to a Spatio-Temporal Graph Attention Network (GAT), modeling stocks as graph nodes with dynamic edge weights $A_{ij,t}$ to capture cross-sectional supply chain linkages, industry spillovers, and dynamic return correlations (Vaswani et al., 2017; Chen et al., 2024).

Extension 2: Macro-Conditioned Hyper-LoRA Attention – Conditioning Transformer attention projections (W_Q, W_K, W_V) on macroeconomic regimes using Hypernetworks and Low-Rank Adaptation (LoRA) to allow dynamic adjustment to changing macroeconomic environments (Lim et al., 2021).

Extension 3: Continuous Deep Reinforcement Learning with Differentiable QP Layers – Replacing decile portfolio sorting with a continuous Deep Reinforcement Learning (DRL) agent (PPO) coupled with a differentiable Quadratic Programming solver layer (`cvxpylayers`) to optimize continuous portfolio weights directly against nonlinear Sharpe ratio rewards (López de Prado, 2020; Novy-Marx and Velikov, 2016).

Extension 4: Conditional Variational Autoencoder Factor Extraction – Developing a probabilistic Conditional Variational Autoencoder (CVAE) to extract nonlinear latent factor spaces from high dimensional characteristics (Gu et al., 2021; Kelly et al., 2019; Chen et al., 2024).

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